CSE260 Fall 2025

Assignment 01

This assignment must be hand-written. Show ALL steps in ALL questions
[You do not need to submit the UNGRADED tasks]

1. Convert the following numbers to the equivalent given base.
   (a) $(114.015625)_{10}$ to binary
   (b) $(5463)_{7}$ to base 13
   (c) $(593.0618)_{10}$ to Excess 4
   (d) $(11010101011110101.01110101011)_{2}$ to hexadecimal
   (e) $(10110100110)_{1's}$ to base 10
   (f) $(00110111101)_{2's}$ to base 10
   (g) $(4463.0712)_{8}$ to binary
   (h) $(1100001101101011010.01011001)_{\text{Excess 3}}$ to Decimal
   (i) $(10110100110)_{2's}$ to base 10
   (j) $(67)_{10}$ to base 6
   (k) $(100110.1101)_{4}$ to base 9

2. Perform addition, subtraction and multiplication for the pair of following numbers separately. You should not change the given base
$(2FB4)_{18}$ and $(8B)_{18}$

3. Perform the following division without changing the given base. Show the resulting quotient and the remainder.
   (a) $(101224)_{6} \div (112)_{6}$
   (b) $(33231200)_{4} \div (30)_{4}$ **[ungraded]**

4. **[Ungraded]** You are a computer enthusiast and want to buy two 8 GB DDR4 RAMs. Each RAM costs $(1E)_{16}$ dollars. You also want to buy a graphics card RTX4070Ti which costs $(10100010100)_{2}$ dollars. However, you don't have that much money and are afraid to ask your parents about it. Suddenly, one of your generous friends agreed to give you the money you need. He decided to give you $(2734)_{8}$ dollars. How much will you have left after buying those components? (Show the answer in decimal)

5. Add +45 with -93 using 7-bit 1's complement number system. Justify if there is an overflow. [Show calculations]

6. Add +91 and -35 using 8-bit 2's complement number system. Justify if there is an overflow. [Show calculations]